

# Diffusion Models and Uncertainty Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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**Notation.** Throughout this problem set we adopt the notation of the course slides:  $\mathbf{x}_0$  is a clean data sample;  $\mathbf{x}_t$  is the noised sample at timestep  $t$ ;  $\beta_t$  is the noise schedule;  $\alpha_t = 1 - \beta_t$ ;  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ ;  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  denotes standard Gaussian noise;  $\epsilon_\theta(\mathbf{x}_t, t)$  is the learned noise-prediction network;  $\nabla_{\mathbf{x}} \log p(\mathbf{x})$  denotes the score function.

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## PART A: CONCEPTUAL QUESTIONS

### A1. Score Function Interpretation

**Question.** The score function is defined as  $s(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$ .

- (a) Explain geometrically what the score vector tells us at a given point  $\mathbf{x}$ .
- (b) Why does the score not require knowledge of the normalizing constant of  $p(\mathbf{x})$ ?

### A2. Noise Prediction vs. $x_0$ -Prediction vs. Score

**Question.** DDPM can be parameterized to predict: (i) the injected noise  $\epsilon$ , (ii) the clean sample  $\mathbf{x}_0$ , or (iii) the score  $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)$ .

- (a) Show that these three parameterizations are algebraically equivalent by expressing each in terms of the others using the forward-process relation  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ .

### A3. Diffusion vs. Kalman Filtering

**Question.** Both diffusion models and Kalman filters involve iterative Gaussian updates.

- (a) Identify at least two structural similarities between the DDPM reverse process and the Kalman filter update step.
  - (b) Identify at least two key differences (e.g. linearity, learned parameters, direction of time, role of observations).
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## PART B: MATHEMATICAL DERIVATIONS

### B1. Closed-Form Forward Process

**Problem.** The DDPM forward process defines

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad \alpha_t = 1 - \beta_t.$$

- (a) By induction (or by direct substitution), derive the closed-form marginal

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

- (b) Verify that the mean coefficient squared plus the variance equals 1 for all  $t$ . Why is this property important?
- (c) What does the distribution  $q(\mathbf{x}_T \mid \mathbf{x}_0)$  converge to as  $T \rightarrow \infty$  under standard schedules?

### B2. Reverse Process Mean as a Denoiser

**Problem.** The ancestral sampling update with  $\epsilon$ -prediction is

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

- (a) Show that, when  $\epsilon_\theta(\mathbf{x}_t, t) = \epsilon$  (oracle noise predictor), the deterministic part of the update recovers a point on the line between  $\mathbf{x}_t$  and the clean sample  $\mathbf{x}_0$ .
- (b) Interpret the deterministic part as a denoising step: what fraction of the noise does it remove?
- (c) What is the role of the stochastic term  $\sigma_t \mathbf{z}$ ? What happens if  $\sigma_t = 0$  for all  $t$ ?

### B3. Denoising Score Matching

**Problem.** Consider corrupting clean data by Gaussian noise:  $\tilde{\mathbf{x}} = \mathbf{x} + \sigma \epsilon$ ,  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ .

- (a) Write the conditional distribution  $q(\tilde{\mathbf{x}} \mid \mathbf{x})$  and derive  $\nabla_{\tilde{\mathbf{x}}} \log q(\tilde{\mathbf{x}} \mid \mathbf{x})$ .
- (b) Show that the denoising score matching objective  $\mathbb{E}_{\mathbf{x}, \epsilon} [\|s_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q(\tilde{\mathbf{x}} \mid \mathbf{x})\|^2]$  is equivalent (up to a constant) to  $\mathbb{E} [\|s_\theta(\tilde{\mathbf{x}}) + \epsilon/\sigma\|^2]$ .
- (c) Explain the conceptual bridge: how does minimizing this objective train the network to approximate the score of the noisy data distribution?

### B4. From ELBO to the Simplified Loss

**Problem.** The DDPM variational lower bound is

$$\mathcal{L}_{\text{VLB}} = D_{\text{KL}}(q(\mathbf{x}_T \mid \mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t=2}^T D_{\text{KL}}(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0 \mid \mathbf{x}_1).$$

- (a) Explain the role of each of the three terms.
- (b) The posterior  $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$  is a Gaussian. State its mean and variance without derivation (you may cite Ho et al., 2020).
- (c) Show qualitatively how the KL term for each  $t$  reduces to a weighted MSE between the true and predicted denoising directions, leading to  $\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2]$ .

## B5. Langevin Dynamics: Stationary Distribution

**Problem.** Consider the Langevin update

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \eta \nabla_{\mathbf{x}} \log p(\mathbf{x}_k) + \sqrt{2\eta} \mathbf{z}_k, \quad \mathbf{z}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

- (a) Identify the deterministic and stochastic components. What would happen if the noise term were removed (i.e.  $\mathbf{z}_k = \mathbf{0}$ )?
  - (b) Under what conditions on  $\eta$  does this Markov chain converge to the target distribution  $p(\mathbf{x})$ ?
  - (c) Show that for  $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$  the drift term  $\eta \nabla_{\mathbf{x}} \log p(\mathbf{x}_k)$  is a linear function of  $\mathbf{x}_k$  and compute it explicitly.
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## PART C: PARAMETRIC / WHAT-IF ANALYSIS

### C1. Effect of the Number of Diffusion Steps $T$

**Question.**

- (a) If  $T$  is too small (e.g.  $T = 10$ ), what goes wrong with the forward process? What about the reverse process?
- (b) If  $T$  is very large (e.g.  $T = 10000$ ), what are the computational drawbacks?
- (c) How do fast-sampling methods (DDIM, DPM-Solver) mitigate the large- $T$  cost without re-training the network?

### C2. Latent Diffusion: Effect of Compression

**Question.** In latent diffusion, data  $\mathbf{x} \in \mathbb{R}^D$  is encoded to  $\mathbf{z} \in \mathbb{R}^d$  with  $d \ll D$ .

- (a) What is the computational benefit of running diffusion in  $\mathbb{R}^d$  instead of  $\mathbb{R}^D$ ?
  - (b) What information might be lost in the compression? How does this affect the quality of generated samples?
  - (c) If the autoencoder is trained separately (frozen), can the diffusion model compensate for autoencoder artifacts? Why or why not?
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## PART D: ENGINEERING APPLICATIONS

### D1. Time-Series Forecasting with Diffusion

**Problem.** You are designing a probabilistic forecaster for electrical load demand. The observed past is  $\mathbf{x}_{\text{past}} \in \mathbb{R}^{48}$  (past 48 hours) and you wish to forecast  $\mathbf{x}_{\text{future}} \in \mathbb{R}^{24}$  (next 24 hours). You model  $p_{\theta}(\mathbf{x}_{\text{future}} \mid \mathbf{x}_{\text{past}})$  using conditional diffusion (CSDI-style).

- (a) Describe how the training procedure works: what is noised, what serves as the condition, and what loss is minimized?
- (b) At inference time, you generate  $S = 100$  sample trajectories. How do you construct a point forecast and a 90% prediction interval from these samples?
- (c) Explain why the sample spread at hour  $t+24$  is typically wider than at hour  $t+1$ . Connect this to the notion of uncertainty growing with the forecast horizon.
- (d) Compare this diffusion-based forecaster to a Kalman-filter-based approach in terms of distributional assumptions, nonlinearity, and computational cost.

## D2. Inverse Problem: Signal Reconstruction from Noisy Measurements

**Problem.** A 1D signal  $\mathbf{x}_0 \in \mathbb{R}^{256}$  is observed through a linear measurement model  $\mathbf{y} = \mathbf{A}\mathbf{x}_0 + \mathbf{n}$ , where  $\mathbf{A} \in \mathbb{R}^{64 \times 256}$  (compressed sensing) and  $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$ .

You have a pre-trained unconditional diffusion model for signals of dimension 256.

- (a) Explain conceptually how the diffusion model acts as a learned prior  $p(\mathbf{x}_0)$  for Bayesian reconstruction.
- (b) Describe a posterior-sampling strategy: at each reverse step, how can you incorporate the measurement likelihood  $p(\mathbf{y} \mid \mathbf{x}_0)$  to guide the reverse chain toward measurement-consistent solutions?
- (c) You draw  $S = 50$  posterior samples  $\{\hat{\mathbf{x}}_0^{(s)}\}_{s=1}^{50}$ . Define a per-element uncertainty map and explain how it could be used to flag unreliable reconstructed components.
- (d) If  $\sigma_n$  increases (noisier measurements), how should the sample diversity change, and why?

## D3. DiffusionDet-Style Object Detection

**Problem.** DiffusionDet treats object detection as iterative denoising of bounding-box coordinates. Let  $\mathbf{b}_0 = (x_c, y_c, w, h)$  denote a ground-truth box.

- (a) During training,  $\mathbf{b}_0$  is corrupted by  $\mathbf{b}_t = \sqrt{\bar{\alpha}_t} \mathbf{b}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$ . Why is it reasonable to apply the same DDPM forward process to box coordinates rather than to pixel intensities?
- (b) At inference, the model starts from random boxes  $\mathbf{b}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  and iteratively refines them. If you run  $S$  independent reverse chains, how can the spread of the resulting boxes quantify detection uncertainty?
- (c) Propose a criterion for using this detection uncertainty to decide whether a predicted detection is “reliable” enough to act upon in a safety-critical system (e.g. autonomous driving).

## D4. Multimodal Conditioning: Sensor Fusion

**Problem.** Consider a scenario where you want to reconstruct a radar image  $\mathbf{x}_0$  conditioned on two modalities: (i) a coarse satellite image  $\mathbf{c}_1$  and (ii) a set of sparse ground-station measurements  $\mathbf{c}_2$ . You train a conditional diffusion model  $p_\theta(\mathbf{x}_0 \mid \mathbf{c}_1, \mathbf{c}_2)$ .

- (a) Describe how you would incorporate both conditions into the denoiser  $\epsilon_\theta(\mathbf{x}_t, t, \mathbf{c}_1, \mathbf{c}_2)$ . Discuss at least two architectural options (e.g. concatenation, cross-attention).

- (b) If  $\mathbf{c}_2$  is unavailable at test time, how can classifier-free guidance (with random dropout of each condition during training) allow graceful degradation?
- (c) How would you evaluate whether the model’s uncertainty (sample spread) is well-calibrated with respect to the true reconstruction error?

## D5. Diffusion for Signal Inpainting

**Problem.** You observe a biomedical signal  $\mathbf{x} \in \mathbb{R}^{512}$  with a contiguous block of 64 missing samples (e.g. due to sensor dropout). A mask  $\mathbf{M} \in \{0, 1\}^{512}$  indicates observed ( $M_i = 1$ ) vs. missing ( $M_i = 0$ ) entries.

- (a) Describe the repaint-style inference algorithm: at each reverse step  $t$ , how are the observed values re-imposed?
- (b) If the missing block is short (e.g. 8 samples), would you expect the sample diversity for the imputed region to be large or small? Why?
- (c) If the missing block is long (e.g. 128 samples), how does the diversity change? Relate this to the concept of aleatoric uncertainty.
- (d) How does this approach differ from classical linear interpolation or autoregressive imputation?